

[Predicting Default Clients for Taiwan Bank]

GROUP NAME

GROUP 7

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BUSINESS QUESTION

Can the Taiwanese Bank predict whether a client will default payment next month?

PROBLEM TYPE - Classification - The target variable is binary & we're predicting a categorical outcome (whether a client will default =1, or not =0)

TARGET VARIABLE

default.payment.next.month i.e., 1 -defaulter, 0-none]

WHY IT MATTERS FOR TAIWAN BANK

This model supports proactive risk management, allowing the bank to minimize bad debts by adjusting credit limits, freezing high-risk accounts and offering intervention strategies before an official default occurs

Majority-class baseline: a model that always predicts the most common class scores [77.9%] accuracy — without learning anything. Your model must beat this.

SPEAKER GUIDE: State the question clearly, explain the business cost of being wrong, then define your target variable. ~2 min.

DATASET FACTS

Source: Default of Credit Card Clients (Taiwan)

Rows: 30,000 clients

Columns: 25

Missing: No missing values

FEATURES USED

List: [default.payment.next.month, limit_bal]

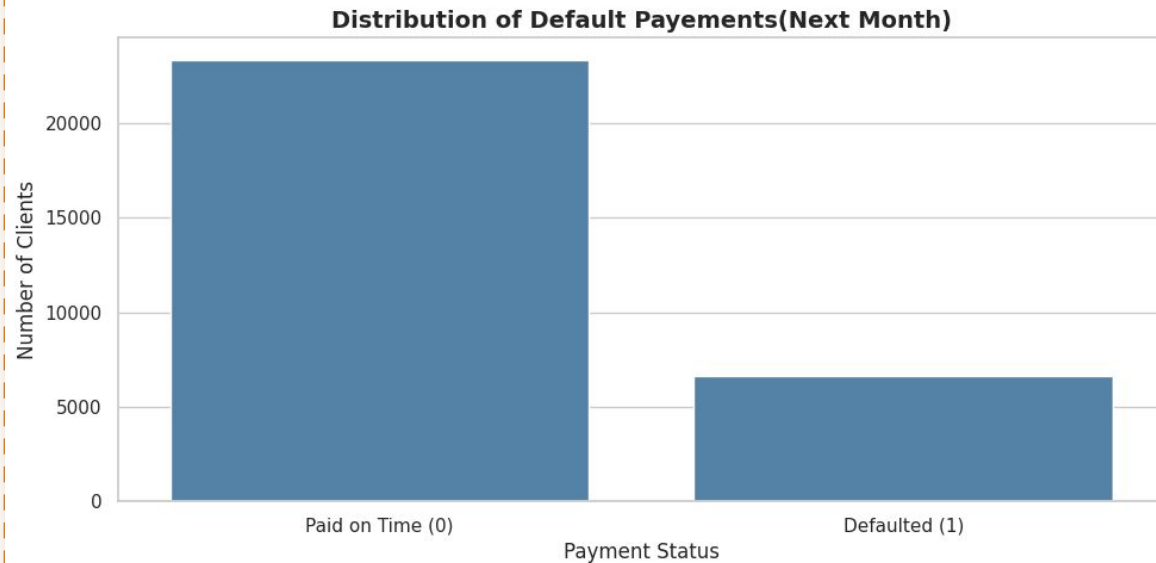
YOUR EDA CHART

Chart insight: Most of the clients in the Taiwanese Bank are non-defaulters.

1

Prepare

```
X =  
pay_0,pay_amt1,bill_amt1,pay_2,limit_bal  
· y = default.payment.next.month
```

2

Split

```
80% train / 20%  
test ·  
random_state=42
```

3

Fit

```
model.fit(X_train  
, y_train)
```

4

Predict

```
y_pred =  
model.predict(X_t  
est)
```

5

Evaluate

Accuracy+F1 on TEST SET only

OUR ALGORITHM

LogisticRegression —

it's simple, fast and provides a good benchmark for comparing more advanced models.

KEY DESIGN DECISION

- *We added class_weight = "balanced"*

Our dataset was highly imbalanced and focusing on the majority class giving us an F1 score of 0.

SPEAKER GUIDE: Walk the 5 steps, state your algorithm, then explain the one decision that most shaped your results.

Train: [24,000] rows · Test: [6,000] rows

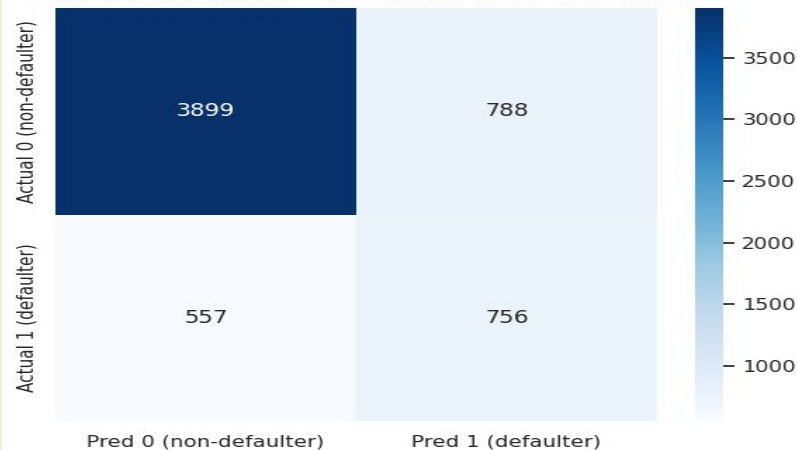
MODEL COMPARISON

Model	Accuracy Score	f1 Score
Majority baseline	[0.7812]	[0.0000]
Logistic Regression	[0.7103]	[0.4824]
Decision Tree Classifier	[0.7758]	[0.5292]

KEY RESULT:

Our improved model achieved 77.58% accuracy (F1=0.5292) — a 0.0655 point improvement over the 71.03% logistic regression baseline.

Confusion Matrix — Decision Tree Classifier



SPEAKER GUIDE: Show table → point to improvement → explain the chart → name the most common error type and its business cost.

MAIN RECOMMENDATION FOR TAIWANESE BANK

The Taiwanese Bank should not rely solely on the ML Model because the model still misses many actual default cases. The bank should combine the model's outputs with additional customer data—specifically customer income and employment status—or incorporate manual reviews to better manage risk and minimize bad debt.

1

Although the model improves identification of customers who are likely to default compared with majority-class baseline, it still misses many default cases and has slightly lower overall accuracy. Therefore, the Taiwan Bank should use the model together with additional information such as income and employment status before making final lending decisions.

2

The Decision Tree improved the F1-score by identifying more customers at risk of default, although its overall accuracy was slightly lower than the baseline. Therefore, Taiwan Bank should use the model to support credit risk assessment alongside other customer information.

3

The model shows that repayment status from earlier months are the strongest predictors of credit card default; therefore, Taiwan Bank should monitor customers with past payment delays and take preventive action before defaults occur.

NEXT STEP:

If we had more time/ data we would collect additional information like customer income, employment status and debt-to-income ratios then train an advanced model like Random Forests to accurately catch the missing default cases.